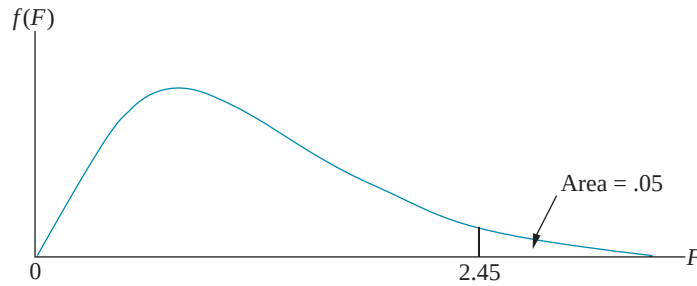


FIGURE 8.4

Critical value of F for $\alpha = .05$, $df_1 = 4$, and $df_2 = 120$



This procedure can be generalized (and simplified) with only slight modifications in the formulas to test the equality of t (where t is an integer equal to or greater than 2) population means from normal populations with a common variance σ^2 . Random samples of sizes $n_1, n_2, \dots, n_t$ are drawn from the respective populations. We then compute the sample means and variances. The null hypothesis $\mu_1 = \mu_2 = \dots = \mu_t$ is tested against the alternative that at least one of the population means is different from the others.

Before presenting the generalized test procedure, we introduce the notation to be used in the formulas for s_B^2 and s_W^2 .

completely randomized design

The experimental setting in which a random sample of observations is taken from each of t different populations is called a **completely randomized design**. Consider a completely randomized design in which four observations are obtained from each of the five populations. If we let y_{ij} denote the j th observation from population i , we could display the sample data for this completely randomized design as shown in Table 8.5. Using Table 8.5, we can introduce notation that is helpful when performing an **analysis of variance (AOV)** for a completely randomized design.

analysis of variance

Notation Needed for the AOV of a Completely Randomized Design

- y_{ij} : The j th sample observation selected from population i . For example, y_{23} denotes the third sample observation drawn from population 2.
- n_i : The number of sample observations selected from population i . In our data set, n_1 , the number of observations obtained from population 1, is 4. Similarly, $n_2 = n_3 = n_4 = n_5 = 4$. However, it should be noted that the sample sizes need not be the same. Thus, we might have $n_1 = 12$, $n_2 = 3$, $n_3 = 6$, $n_4 = 10$, and so forth.
- n_T : The total sample size; $n_T = \sum n_i$. For the data given in Table 8.5, $n_T = n_1 + n_2 + n_3 + n_4 + n_5 = 20$.
- $\bar{y}_i$: The average of the n_i sample observations drawn from population i , $\bar{y}_i = \sum_j y_{ij}/n_i$.
- $\bar{y}_{..}$: The average of all sample observations; $\sum_i \sum_j y_{ij}/n_T$.

TABLE 8.5

Summary of sample data for a completely randomized design

Population	Data				Mean
1	y_{11}	y_{12}	y_{13}	y_{14}	$\bar{y}_{1\cdot}$
2	y_{21}	y_{22}	y_{23}	y_{24}	$\bar{y}_{2\cdot}$
3	y_{31}	y_{32}	y_{33}	y_{34}	$\bar{y}_{3\cdot}$
4	y_{41}	y_{42}	y_{43}	y_{44}	$\bar{y}_{4\cdot}$
5	y_{51}	y_{52}	y_{53}	y_{54}	$\bar{y}_{5\cdot}$

With this notation, it is possible to establish the following algebraic identities. (Although we will use these results in later calculations for s_W^2 and s_B^2 , the proofs of these identities are beyond the scope of this text.) Let s_T^2 be the sample variance of the n_T measurements y_{ij} . We can measure the variability of the n_T sample measurements y_{ij} about the overall mean $\bar{y}_{..}$ using the quantity

$$\text{TSS} = \sum_{i=1}^t \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{..})^2 = (n_T - 1)s_T^2$$

total sum of squares

This quantity is called the **total sum of squares** (TSS) of the measurements about the overall mean. The double summation in TSS means that we must sum the squared deviations for all rows (i) and columns (j) of the one-way classification.

It is possible to partition the total sum of squares as follows:

$$\sum_{i,j} (y_{ij} - \bar{y}_{..})^2 = \sum_{i,j} (y_{ij} - \bar{y}_{i.})^2 + \sum_i n_i (\bar{y}_{i.} - \bar{y}_{..})^2$$

The first quantity on the right side of the equation measures the variability of an observation y_{ij} about its sample mean $\bar{y}_{i.}$. Thus,

$$\text{SSW} = \sum_{i,j} (y_{ij} - \bar{y}_{i.})^2 = (n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 + \cdots + (n_t - 1)s_t^2$$

within-sample sum of squares

is a measure of the *within-sample* variability. SSW is referred to as the **within-sample sum of squares** and is used to compute s_W^2 .

The second expression in the total sum of squares equation measures the variability of the sample means $\bar{y}_{i.}$ about the overall mean $\bar{y}_{..}$. This quantity, which measures the variability *between* (or among) the sample means, is referred to as the **sum of squares between samples** (SSB) and is used to compute s_B^2 .

between-sample sum of squares

$$\text{SSB} = \sum_i n_i (\bar{y}_{i.} - \bar{y}_{..})^2$$

Although the formulas for TSS, SSW, and SSB are easily interpreted, they are not easy to use for calculations. Instead, we recommend using a computer software program.

An analysis of variance for a completely randomized design with t populations has the following null and alternative hypotheses:

$$H_0: \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_t \text{ (i.e., the } t \text{ population means are equal)}$$

$$H_a: \text{At least one of the } t \text{ population means differs from the rest.}$$

The quantities s_B^2 and s_W^2 can be computed using the shortcut formulas

$$s_B^2 = \frac{\text{SSB}}{t - 1} \quad s_W^2 = \frac{\text{SSW}}{n_T - t}$$

where $t - 1$ and $n_T - t$ are the degrees of freedom for s_B^2 and s_W^2 , respectively.

mean square

Historically, people have referred to a sum of squares divided by its degrees of freedom as a **mean square**. Hence, s_B^2 is often called the *mean square between samples* and s_W^2 , the *mean square within samples*. The quantities are the mean squares because they both are averages of squared deviations. There are only $n_T - t$ linearly independent deviations $(y_{ij} - \bar{y}_{i.})$ in SSW because $\sum_j (y_{ij} - \bar{y}_{i.}) = 0$ for each of the t samples. Hence, we divide SSW by $n_T - t$ and not n_T . Similarly, there are only $t - 1$ linearly independent deviations $(\bar{y}_{i.} - \bar{y}_{..})$ in SSB, because $\sum_i n_i (\bar{y}_{i.} - \bar{y}_{..}) = 0$. Hence, we divide SSB by $t - 1$.

The null hypothesis of equality of the t population means is rejected if

$$F = \frac{s_B^2}{s_W^2}$$

exceeds the tabulated value of F for $\alpha = \alpha$, $df_1 = t - 1$, and $df_2 = n_T - t$.

AOV table

After we complete the F test, we then summarize the results of a study in an *analysis of variance table*. The format of an **AOV table** is shown in Table 8.6. The AOV table lists the sources of variability in the first column. The second column lists the sums of squares associated with each source of variability. We showed that the total sum of squares (TSS) can be partitioned into two parts, so SSB and SSW must add up to TSS in the AOV table. The third column of the table gives the degrees of freedom associated with the sources of variability. Again, we have a check; $(t - 1) + (n_T - t)$ must add up to $n_T - 1$. The mean squares are found in the fourth column of Table 8.6, and the F test for the equality of the t population means is given in the fifth column.

TABLE 8.6

An example of an AOV table for a completely randomized design

Source	Sum of Squares	Degrees of Freedom	Mean Square	F Test
Between samples	SSB	$t - 1$	$s_B^2 = \text{SSB}/(t - 1)$	s_B^2/s_W^2
Within samples	SSW	$n_T - t$	$s_W^2 = \text{SSW}/(n_T - t)$	
Totals	TSS	$n_T - 1$		

EXAMPLE 8.1

A horticulturist was investigating the phosphorus content of tree leaves from three different varieties of apple trees (1, 2, and 3). Random samples of five leaves from each of the three varieties were analyzed for phosphorus content. The data are given in Table 8.7. Use these data to test the hypothesis of equality of the mean phosphorus levels for the three varieties. Use $\alpha = .05$.

TABLE 8.7

Phosphorus content of leaves from three different trees

Variety	Phosphorus Content						Sample Sizes	Means	Variances
1	.35	.40	.58	.50	.47		5	0.460	.00795
2	.65	.70	.90	.84	.79		5	0.776	.01033
3	.60	.80	.75	.73	.66		5	0.708	.00617
Total							15	0.648	

Solution The null and alternative hypotheses for this example are

$$H_0: \mu_1 = \mu_2 = \mu_3$$

H_a : At least one of the population means differs from the rest.

The sample sizes are $n_1 = n_2 = n_3 = 5$, which yields $n_T = 15$. Using the sample means and sample variances, the sum of squares within and between are

$$\begin{aligned} \text{SSB} &= \sum_{i=1}^3 n_i(\bar{y}_i - \bar{y}_{..})^2 = 5(.46 - .648)^2 + 5(.776 - .648)^2 + 5(.708 - .648)^2 \\ &= .277 \end{aligned}$$

and

$$\begin{aligned} SSW &= \sum_{i=1}^3 (n_i - 1)s_i^2 = (5 - 1)(.00795) + (5 - 1)(.01033) + (5 - 1)(.00617) \\ &= .0978 \end{aligned}$$

Finally, $TSS = SSB + SSW = .277 + .0978 = .3748$.

The AOV table for these data is shown in Table 8.8. The critical value of $F = s_B^2/s_W^2$ is 3.89, which is obtained from Table 8 in the Appendix for $\alpha = .05$, $df_1 = 2$, and $df_2 = 12$. Because the computed value of F , 17.25, exceeds 3.89, we reject the null hypothesis of equality of the mean phosphorus content for the three varieties. It appears from the data that the mean for variety 1 is smaller than the means for varieties 2 and 3. We will develop techniques to confirm this observation in Chapter 9.

TABLE 8.8
AOV table for the data
for Example 8.1

Source	Sum of Squares	Degrees of Freedom	Mean Square	F Test
Between samples	.277	2	.277/2 = .138	.138/.008 = 17.25
Within samples	.0978	12	.0978/12 = .008	
Totals	.3748	14		

EXAMPLE 8.2

A clinical psychologist wished to compare three methods for reducing hostility levels in university students, and used a certain test (HLT) to measure the degree of hostility. A high score on the test indicated great hostility. The psychologist used 24 students who obtained high and nearly equal scores in the experiment. Eight were selected at random from among the 24 problem cases and were treated with method 1. Seven of the remaining 16 students were selected at random and treated with method 2. The remaining nine students were treated with method 3. All treatments were continued for a one-semester period. Each student was given the HLT test at the end of the semester, with the results shown in Table 8.9. Use these data to perform an analysis of variance to determine whether there are differences among mean scores for the three methods. Use $\alpha = .05$.

TABLE 8.9
HLT test scores

Method	Test Scores								Mean	Standard Deviation	Sample Size
1	96	79	91	85	83	91	82	87	86.750	5.625	8
2	77	76	74	73	78	71	80		75.571	3.101	7
3	66	73	69	66	77	73	71	70	71.000	3.674	9

Solution The null and alternative hypothesis are

$$H_0: \mu_1 = \mu_2 = \mu_3$$

H_a : At least one of the population means differs from the rest.

For $n_1 = 8$, $n_2 = 7$, and $n_3 = 9$, we have a total sample size of $n_T = 24$. Using the sample means given in the table, we compute the overall mean of the 24 data values:

$$\begin{aligned}\bar{y}_{..} &= \sum_{i=1}^3 n_i \bar{y}_i / n_T = (8(86.750) + 7(75.571) + 9(71.000)) / 24 = 1,861.997 / 24 \\ &= 77.5832\end{aligned}$$

Using this value along with the means and standard deviations in Table 8.9, we can compute the three sums of squares as follows:

$$\begin{aligned}\text{SSB} &= \sum_{i=1}^3 n_i (\bar{y}_i - \bar{y}_{..})^2 = 8(86.750 - 77.5832)^2 + 7(75.571 - 77.5832)^2 \\ &\quad + 9(71 - 77.5832)^2 = 1,090.6311\end{aligned}$$

and

$$\begin{aligned}\text{SSW} &= \sum_{i=1}^3 (n_i - 1) s_i^2 = (8 - 1)(5.625)^2 + (7 - 1)(3.101)^2 + (9 - 1)(3.674)^2 \\ &= 387.1678\end{aligned}$$

Finally, $\text{TSS} = \text{SSB} + \text{SSW} = 1,090.6311 + 387.1678 = 1,477.80$. The AOV table for these data is given in Table 8.10.

TABLE 8.10
AOV table for data
of Example 8.2

Source	SS	df	MS	<i>F</i>	<i>p</i> -value
Between samples	1,090.6311	2	545.316	545.316/18.4366 = 29.58	<.001
Within samples	387.1678	21	18.4366		
Totals	1,477.80	23			

The critical value of F is obtained from Table 8 in the Appendix for $\alpha = .05$, $df_1 = 2$, and $df_2 = 21$; this value is 3.47. Because the computed value of F is 29.57, which exceeds the critical value 3.47, we reject the null hypothesis of equality of the mean scores for the three methods of treatment. We can only place an upper bound on the p -value because the largest value in Table 8 for $df_1 = 2$ and $df_2 = 21$ is 9.77, which corresponds to a probability of .001. Thus, there is a very strong rejection of the null hypothesis. From the three sample means, we observe that the mean for method 1 is considerably larger than the means for methods 2 and 3. The researcher would need to determine whether all three population means differ or the means for methods 2 and 3 are equal. Also, we may want to place confidence intervals on the three method means and on their differences; this would provide the researcher with information concerning the degree of differences in the three methods. In the next chapter, we will develop techniques to construct these types of inferences. Computer output shown here is consistent with the results we obtained. In the computer printout, note that the names for the sum of squares are not given as between and within. The between sum of squares is labeled by the population name, in this example, Method. The within sum of squares is often labeled as the error sum of squares.

General Linear Models Procedure

Class Level Information

Class	Levels	Values
METHOD	3	1 2 3

Number of observations in data set = 24

Dependent Variable: SCORE

Source	DF	Sum of Squares	F Value	Pr > F
Model	2	1090.61904762	29.57	0.0001
Error	21	387.21428571		
Corrected Total	23	1477.83333333		

EXERCISES Applications

Ag.

8.1 A large laboratory has four types devices used to determine the pH of soil samples. The laboratory wants to determine whether there are differences in the average readings given by these devices. The lab uses 24 soil samples having known pH in the study, and randomly assigns six of the samples to each device. The soil samples are tested and the response recorded is the difference between the pH reading of the device and the known pH of the soil. These values, along with summary statistics, are given in the following table.

Device	Sample						Sample Size	Mean	Standard Deviation
	1	2	3	4	5	6			
A	-.307	-.294	.079	.019	-.136	-.324	6	-.1605	.1767
B	-.176	.125	-.013	.082	.091	.459	6	.0947	.2091
C	.137	-.063	.240	-.050	.318	.154	6	.1227	.1532
D	-.042	.690	.201	.166	.219	.407	6	.2735	.2492

- Based on your intuition, is there evidence to indicate any difference among the mean differences in pH readings for the four devices?
- Run an analysis of variance to confirm or reject your conclusion of part (a). Use $\alpha = .05$.
- Compute the p -value of the F test in part (b).
- What conditions must be satisfied for your analysis in parts (b) and (c) to be valid?
- Suppose the 24 soil samples have widely different pH values. What problems may occur by simply randomly assigning the soil samples to the different devices?

Bus.

8.2 A cigarette manufacturer has advertised that it has developed a new brand of cigarette, LowTar, that has a lower average tar content than the major brands. To evaluate this claim, a consumer testing agency randomly selected 100 cigarettes from each of the four leading brands of cigarettes and 100 from the new brand. The tar content (milligrams) of the cigarettes gave the following results: